Generalized Evolution Theory: The Accelerated Evolution of Human Civilization in the Noetic Ecology

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Abstract

While academia largely agrees that human biological evolution has slowed significantly—even stagnated—due to advances in medical technology, the complexity of civilization systems has surged exponentially. From technological breakthroughs to cultural shifts, this trajectory speaks volumes to the fact that evolutionary momentum has not diminished. This contradiction stems from the framework limitations of narrow-sense evolution theory: it confines evolution to the material world, overlooking information's core role as an evolutionary carrier in the information realm. We put forward the Generalized Evolution Theory, expanding evolution's scope from material systems to noetic systems (cognitive-informational systems) driven by logical self-reference and self-organization tendencies. Through axiomatic deduction, we argue that noetic genes (symbols, narratives, cognitive structures, etc.), fueled by logical self-reference and self-organization, exhibit high-speed mutation and short iteration cycles—ultimately driving civilization's accelerated evolution. Beyond unifying biological, cultural, and technological evolution, the Generalized Evolution Theory offers a new framework for diagnosing civilizational health.

Keywords: Generalized Evolution Theory; Noetic Gene; Accelerated Evolution; Collaborative Cognition

Core Concepts at a Glance

To facilitate understanding, we briefly define key theoretical constructs repeated throughout the paper. Their full elaboration appears in Chapter 3.

Notably, Noetic Ecology:

The theoretical backbone of this work, aiming to build a unified framework for understanding the evolution of human cognition, society, and culture. It does not seek to be a closed "final truth" resolving all questions, but rather an open "cognitive operating system." Through a self-consistent axiomatic system, it provides a coherent, derivable, and testable approach to addressing issues ranging from individual meaning-making to civilizational forms.
The term 'noetic' (from Greek 'nous', meaning mind or intellect) describes systems that exhibit both cognitive processing and informational organization. In this framework, it bridges the conceptual gap between purely material and purely informational approaches to evolution.

Indeed, Dynamic Principle of Logical Self-Reference (Axiom 0):

Any logical system reaching a certain complexity will develop "self-reference"—the ability to treat itself or its subsets as objects of operation. This spawns a dynamic, self-iterative "logical sub-universe" within the system, whose evolution is the core source of adaptive innovation in response to environmental challenges. Creative intuition, conscious reflection, and even civilizational paradigm shifts are all macroscopic phase transitions emerging from this "logical sub-universe"—forged through internal simulation and iteration when the system faces survival pressure.

Essentially, Material/Noetic Duality (Axiom 1):

The theory's cornerstone. It posits that any system of sufficient complexity exists as a duality: material entity and noetic pattern. For instance, humans are simultaneously unities of material form (biological brain, body) and noetic systems (thoughts, culture, cognitive frameworks).

Crucially, Self-Organization Tendency of Systems (Axiom 2):

Also termed "generalized will to survive" by the author, this is another theoretical cornerstone. It holds that all systems—from galaxies and life to noetic ecology—exhibit an inherent tendency toward self-organization, moving away from thermodynamic equilibrium. This is a universal phenomenon of local entropy reduction in the universe. At the noetic level, this tendency manifests as preserving and expanding one's own low-entropy ordered structure (i.e., noetic state)—an experience we recognize as "will to survive."

Central to this work, Generalized Evolution Theory (Theorem 37):

The paper's core argument. It asserts that evolution is a universal process: complex systems achieve hierarchical leaps through "mutation-selection-inheritance," a scope that naturally extends from biological genes to noetic systems.

Simply put, Symbol:

The "atom" of the noetic world, acting as a carrier of concepts or meanings (e.g., the number "1," the word "freedom").

Specifically, Information:
A static existence with a stable topological structure—symbols arranged according to specific rules (logic, grammar, cultural norms). Examples include the equation "1+1=2," a philosophical treatise, or a database.

At its core, Noesis:
Noesis: In contrast to static 'information,' noesis (operationalized as cognitively active, self-organizing information systems) refers to a dynamic, inherently driven complex information system (e.g., the human brain, an active academic community, or even civilization). It actively absorbs, deconstructs, and reconstructs information to drive innovation.
Its core dynamics stem primarily from:
- Self-organization tendency (Axiom 2): The system's inherent "will to survive"—the instinct to maintain and expand its ordered structure.
- Logical self-reference (Axiom 0): The system's ability to examine and operate on itself, constructing a "logical sub-universe" for simulation and iteration.

Fundamentally, Noetic Gene:
The basic unit of heredity and mutation in complex information systems with self-organization tendencies—specifically, symbolic structures (e.g., a scientific concept, an institutional design, a technical blueprint).

Practically, External Brain Domain:
A technological cognitive environment collectively built by humans (e.g., the Internet, artificial intelligence). As an extension of the noetic system, it drastically accelerates the processing and dissemination of noetic genes.

First-Order Noetic Humans:
Individuals with strong metacognitive abilities—those who can recognize their place in the noetic ecology and resist the obliteration effect of the macro system.

Second-Order Noetic Humans:
The bedrock of civilization. By connecting to collaborative cognitive networks, they achieve cognitive resonance with other noetic humans, emerging with the ability to guide collective meta-consciousness.

Collective Mind:
A macro system blindly emerging from the unconscious interactions of individuals under the old paradigm.

Metacognitive Collective Intelligence:
A macro intelligence with civilizational-level self-observation and guidance capabilities—emerging when second-order noetic humans connect through collaborative cognitive networks.

Logical Sub-Universe:
A dynamic cognitive space internally constructed by complex systems (especially human noetic systems) via logical self-reference. It enables simulation, iteration, and meaning-making, serving as the operational field for creative intuition, philosophical speculation, and paradigm revolutions.

System Parasite:

In any complex social system with limited resources, as the system's structure stabilizes and rigidifies, certain subsystems—"system parasites"—inevitably emerge. Their defining trait is an extreme imbalance between responsibility and benefit: they maximize their own value extraction by systematically distorting rules and shirking their duty to maintain the parent system. Their fundamental harm lies in systematically and continuously hindering the comprehensive development and potential realization of other individuals in the parent system—thus betraying the system's ultimate value as defined in [Axiom 5].

1. Introduction

1.1 The Paradox of Evolutionary Stagnation Hypothesis and Accelerated Reality

Since Darwin's pioneering work, evolution theory **serves** as the cornerstone for understanding biological evolution. Modern biological research—particularly genomic analyses—largely supports a collective view: advances in medical technology and social welfare have significantly weakened natural selection in human societies, leading to a substantial slowdown, even stagnation, of human biological evolution (Hawks et al., 2007; Stearns et al., 2010). This "evolutionary stagnation hypothesis" enjoys broad consensus in academia.
Yet, in stark contrast to this biological "stagnation," human civilization—a complex system—has undergone unprecedented accelerated change. The cycle of technological iteration has shrunk from decades to years, even months (Kurzweil, 2005). Knowledge accumulates exponentially, while social structures and cultural paradigms are restructured at an unparalleled pace. This explosive growth in civilizational complexity poses a core scientific paradox: if the driving force behind evolution has indeed stagnated, how, then, can we account for the macro developmental trend of civilizations—one that is far more directional and accelerating than the randomness of biological evolution?
Clearly, confining evolution to the traditional framework of biological genetic mutation and natural selection fails to provide a coherent explanation for this paradox.

1.2 Limitations of Existing Theories: Shortcomings from Memetics to Cultural Evolution

Scholars have not shied away from transcending the boundaries of biological evolution. Dawkins' (1976) concept of "meme"—analogizing information units in cultural systems to genes—pioneered the field of cultural evolution research. Subsequent cultural evolution theory (Boyd & Richerson, 1985) attempted to model how cultural traits spread, mutate, and are selected through social learning.
Unfortunately, these theoretical frameworks face fundamental predicaments. For one thing, their selection mechanisms remain ambiguous: memetics lacks a clear "selective death" mechanism analogous to natural selection, rendering the active process of "cultural evolution" arbitrary and difficult to quantify. For another, they suffer from a disconnection from material foundations—most existing theories treat cultural evolution as a relatively independent trajectory, failing to ontologically integrate it with underlying biological evolution and technological systems as material entities. Most critically, their explanatory power falls short: they struggle to account for the sharp acceleration of noetic evolution in recent decades—especially the catalytic role of emerging technologies represented by artificial intelligence (AI).
These limitations indicate that what we need is not a patchwork of existing theories, but a paradigmatic leap—a new framework capable of unifying the understanding of evolutionary laws across material, information, biological, and civilizational scales.

1.3 Our Approach: Noetic Ecology and Generalized Evolution Theory

To resolve the aforementioned paradox and limitations, **we introduce** the axiomatic system of "Noetic Ecology" as a theoretical container and propose the "Generalized Evolution Theory" as its core argument. Our core proposition is: human evolution has not stagnated; instead, its main battlefield has shifted from the biological to the noetic level.
This approach is grounded in axioms such as 'Material/Noetic Duality'—where 'noetic' denotes the organized cognitive-informational dimension of complex systems (detailed in Chapter 3), which requires us to view

civilization as a macro system of material-noetic co-evolution. From this perspective, the cohesion of civilization stems from shared meta-narratives and protocols, while its fundamental driving force for evolution lies in the self-organization tendency and logical self-reference ability of the noetic system. From this, **we derive** the "Generalized Evolution Theorem," redefining evolution as a universal dynamic process spanning material and noetic domains.

In this framework, "human evolutionary stagnation" is merely an observational illusion confined to the material world. Once we adopt the perspective of "Material/Noetic Duality," we cannot limit evolution to the single dimension of biological genetic mutation. Instead, evolution should be understood as a complex process of co-evolution between material and noetic states. At present, the evolution of "noetic genes"—units such as symbols, narratives, and cognitive frameworks—has become the dominant factor driving the macro evolution of civilized systems due to their extremely high mutation and selection rates; crucially, we do not regard it as the sole factor.

This judgment is based on the following facts:

- Sustained support from material substrates: As the material carrier of noetic activities, the biological brain continues to evolve slowly but steadily. Its cognitive capabilities form the cornerstone of all noetic activities—without this material foundation, noetic evolution would be a source without water.
- Synchronous expansion of material entities: We know that the claws, teeth, and wings of organisms are "built-in tools" generated under genetic guidance. Why, then, cannot we regard external entities created through intellectual labor as "external tools" designed and manipulated through noetic processes? From this perspective, AI can be seen as an "External Brain Domain" (noetic extension), and its hardware (servers, chips, networks) as the "external material entities" of our civilization. Thus, the material evolution of humans has shifted from relying on slow biological genetic mutation to a new paradigm of rapidly iterating "external material entities" through noetic blueprints.

In this sense: We do not replace material evolution with noetic evolution, but integrate material evolution into a faster feedback loop driven by noetic processes.

1.4 Structure and Contributions of the Paper

This paper aims to systematically elaborate on the Generalized Evolution Theory and demonstrate the mechanisms underlying the accelerated evolution of civilization. Here is how we organize our argument:

- First, Chapter 2 (Literature Review) covers a systematic review of biological evolution, cultural evolution, and related philosophical discussions—clarifying the academic context of this theory and where it transcends existing work.
- Chapter 3 (Theoretical Framework) lays out the axiomatic foundation of Noetic Ecology, formally derives the generalized evolution model, and defines core concepts such as "noetic gene" and "External Brain Domain."
- Next, Chapter 4 (Mechanism Demonstration) explores the internal dynamics of accelerated evolution from three perspectives—strengthening of self-organization tendency, civilizational phase transition model, and amplification effect of logical self-reference—combining qualitative and quantitative analysis.
- Chapter 5 (Case Studies) tests the theory against reality, using two cases: the technological evolution of AI and the rapid spread of specific cultural symbols (e.g., "involution").
- Finally, Chapter 6 (Discussion and Conclusion) summarizes our key findings, addresses potential

criticisms, and elaborates on the profound significance of the Generalized Evolution Theory for understanding the future of civilization and formulating development strategies.

Our key contributions are threefold:

- We propose an axiomatic, unified ontological framework that integrates biological evolution and noetic evolution into the same dynamic system.
- We explicitly define AI as an intrinsic component of human noetic evolution (rather than an external tool), offering a new paradigm for understanding human-machine relations.
- We provide a testable model to explain the phenomenon of civilizational acceleration—particularly through phase transition conditions in "civilizational dynamics" and the "Cognitive Escape Theorem," enabling the quantitative discussion of evolutionary rates.

Through this work, we aim to promote the necessary paradigmatic expansion of evolution theory, providing a more solid and predictive scientific foundation for understanding humanity’s unique trajectory in the universe.

2. Literature Review

2.1 The Paradigmatic Evolution of Evolutionary Thought: From Biology to Culture

Since Darwin (1859) proposed evolution theory, we trace its profound transformation from a discipline-specific concept in biology to an interdisciplinary paradigm. The Modern Synthesis (Huxley, 1942) unified genetics and natural selection, establishing a gene-centric view of evolution. However, this classical framework has increasingly revealed its boundaries when confronting the rapid changes of human society.

2.1.1 Achievements and Limitations of Biological Evolution Theory

For starters, genomic research (Venter, 2007) has confirmed the foundational role of natural selection in shaping human biological traits, uncovering evolutionary traces such as disease resistance and environmental adaptation. Yet a critical limitation emerges from studies led by Hawks et al. (2007) and Stearns et al. (2010): modern medical care and social security systems have significantly reduced the intensity of natural selection, leading to a marked slowdown in phenotypic evolution. This "evolutionary stagnation hypothesis" has become a consensus in contemporary academia.

2.1.2 Breakthroughs and Dilemmas of Cultural Evolution Theory

Dawkins (1976) pioneered cultural evolution research with the concept of "meme" in *The Selfish Gene*, analogizing information units in cultural systems to biological genes. Subsequent Dual Inheritance Theory (Boyd & Richerson, 1985) systematically elaborated on how cultural traits spread through social learning, while Mesoudi et al. (2006) further developed quantitative models for cultural evolution.
Three core dilemmas persist, however:

- First, their selection mechanisms remain ill-defined. Unlike natural selection in biology, cultural evolution lacks clear criteria for "selection," making it difficult to explain why certain cultural traits persist while others fade (Sperber, 1996).
- Second, defining their core units proves challenging. Memes lack the clear boundaries of genes, rendering them hard to define and measure precisely (Blackmore, 1999).
- Third, they remain disconnected from material foundations—failing to explain the dynamic link between cultural evolution and technological innovation (Arthur, 2009).

2.2 The Expansion of Technological Evolution and Information Theory

2.2.1 Technology as an Evolutionary System

In *The Nature of Technology*, Brian Arthur (2009) argues that technology evolves through "combinatorial evolution": new technologies emerge from combinations of existing ones, which then become building blocks for future innovations. This process exhibits distinct evolutionary characteristics—mutation (innovation), selection (market adoption), and inheritance (knowledge accumulation).

2.2.2 The Philosophical Foundation of Information Evolution

Szathmáry and Maynard Smith's (1995) "major evolutionary transitions" theory identifies innovations in information storage and transmission as key drivers of evolution. From genetic codes and animal signals to language and writing, each transition has reshaped the evolutionary landscape. We argue that the "External

Brain Domain"—centered on AI—represents a new major transition, revolutionizing human collective memory, computing power, and linguistic organization.
Even so, these theories leave obvious gaps: technological evolution theory overlooks the role of cognitive frameworks; information philosophy fails to provide operational models for selection mechanisms; and none systematically explores AI as a potential new carrier of evolution. Based on the logical deduction of our theory (detailed in Chapter 3), we propose a testable hypothesis: by carrying, processing, and restructuring noetic genes, AI systems are no longer mere tools but have begun to act as evolutionary carriers. Verifying this hypothesis will require future empirical studies measuring AI-driven cultural mutation rates and selection mechanisms.

2.3 Evolutionary Thought in Systems Science and Complexity Theory

2.3.1 Self-Organization and Evolution

Systems science offers a meta-framework for understanding evolution beyond biology. Prigogine's (1984) dissipative structure theory demonstrates that open systems can spontaneously generate ordered structures when driven by energy flow—a phenomenon known as "self-organization." Subsequent complex systems theories adopted this principle to explain macro-order emergence, from the origin of life to urban development.
Yet most of these theories rely on an analogical concept of "selection" when explaining evolutionary direction and rate, rarely clarifying its underlying dynamics. For example, many models treat "fitness" as an exogenous parameter rather than a product of internal system interactions. In our framework, we aim to provide a more fundamental systemic explanation for "selection." The "generalized will to survive" defined in [Axiom 2] (Self-Organization Tendency of Systems) injects continuous directional momentum into "self-organization." What we call "natural selection" can be ontologically reduced to the statistical outcome of [Axiom 2] under resource constraints, through [system existence and interactions]. It is not a "selector" but a dynamic equilibrium state arising from competition and collaboration between systems to maintain their existence.
This perspective offers a key advantage: it makes "selection" modifiable. In our formal construction (see Section 3.3), the "selection pressure" in both the noetic gene transmission equation and the cognitive escape model stems from the interaction between the intrinsic survival will of system components and external environmental constraints. This avoids teleological assumptions and lays the groundwork for quantifying civilizational evolution rates.

2.3.2 Complex Adaptive Systems Theory

Holland's (1995) Complex Adaptive Systems (CAS) theory posits that systems evolve adaptively by adjusting their behavioral rules through experience accumulation. This framework provides a common language for understanding evolution in economics, ecology, and other fields.
Despite their broad perspective, these systems science theories suffer from two critical shortcomings: they fail to integrate biological, cognitive, and cultural evolution into a single theoretical framework; and they offer metaphors rather than testable, concrete mechanisms.

2.4 Insights from Cognitive Science and Artificial Intelligence

2.4.1 Extended Evolution Theory

Tomasello’s (1999) "cultural origin theory" emphasizes that humans’ unique cultural learning abilities—particularly "intention reading" and "social imitation"—gave rise to cumulative cultural evolution. This perspective places cognitive capabilities at the core of cultural evolution.

2.4.2 Artificial Intelligence as a New Frontier of Evolution

Recent AI research has begun to exhibit evolutionary characteristics: the "mutation-selection" process in deep learning models (LeCun et al., 2015); the ability of large language models to internalize and restructure cultural knowledge (Brown et al., 2020); and the spontaneous emergence of cooperative behavior in multi-agent systems (Leibo et al., 2017).

Yet most existing studies treat AI as a tool rather than an evolutionary agent, failing to recognize its essence as an extension of the human noetic system.

2.5 Theoretical Gaps and Research Opportunities

Synthesizing the existing literature, we pinpoint three key gaps in understanding human evolution:

- An ontological gap: Material and information evolution are still treated as separate domains, lacking a unified ontological framework.
- A mechanism gap: There is no systematic dynamic explanation for the phenomenon of accelerated contemporary evolution.
- An integration gap: AI and other technological systems have not been organically integrated into evolutionary narratives.

Our Research Positioning:

Our work fills these gaps by placing the "Generalized Evolution Theory" at the intersection of these unresolved issues. We do not discard existing research; instead, we weave these scattered insights into a cohesive whole through the axiomatic system of "Noetic Ecology." Specifically:

- We bridge the ontological gap with "Material/Noetic Duality," establishing a unified ontological foundation for biological and noetic evolution and viewing civilization as a system of material-noetic co-evolution.
- We explain the acceleration mechanism with "self-organization tendency" and the "civilizational phase transition model," dynamically linking evolutionary rate to variables such as the proportion of metacognitive individuals and the maturity of collaborative protocols.
- We integrate biological, cultural, and technological evolution through the "Generalized Evolution Theory" framework. Within this framework, biological genes drive material evolution, while noetic genes drive noetic evolution (encompassing culture and technology), forming a feedback loop through interfaces such as the "External Brain Domain" to constitute a complete picture of civilizational evolution.

In the next chapter, we will elaborate on the logical foundation and core components of this theoretical framework.

3. Theoretical Framework: Noetic Ecology and Generalized Evolution Theory

3.1 The Axiomatic Foundation of Noetic Ecology

We anchor our theoretical construction in the axiomatic system of *Noetic Ecology*. Below are the absolute premises for deduction:

- Notably, **Axiom 0 (Dynamic Principle of Logical Self-Reference)**: Any logical system reaching a certain complexity will emerge with self-referential capacity. This ability allows the system to treat itself or its subsets as operational objects, fostering a dynamic "logical sub-universe" within. It serves as the source of consciousness, metacognition, and creative thinking—and more critically, the core driver behind the accelerated mutation and selection of noetic genes.
- Essentially, Axiom 1 (Material/Noetic Duality): Any system of sufficient complexity manifests as a hybrid structure of material entity and noetic pattern (the cognitively organized information regime). The material state acts as the carrier and energy foundation (e.g., biological brains or hardware), while the noetic state encompasses the patterns, rules, and representations of information organization within and between systems (e.g., cognitive frameworks or cultural narratives). This axiom forms our ontological bedrock: evolutionary research must encompass the co-evolution of both material and noetic dimensions.
- Crucially, **Axiom 2 (Self-Organization Tendency of Systems)**: All systems exhibit an inherent tendency toward self-organization, moving away from thermodynamic equilibrium. At the noetic level, this tendency manifests as the drive to maintain and expand one's own low-entropy ordered structure (will to survive)—the fundamental engine of evolution. It propels biological adaptation and fuels noetic innovation alike.
- Concretely, **Axiom 4 (Symbols as Noetic Genes)**: Within the noetic ecology, all carriers of meaning—including concepts, narratives, images, and institutional models—qualify as symbols. They function analogous to genes in the biological realm, serving as the basic units of cultural heredity, mutation, and selection.

These axioms collectively ensure the logical consistency and universality of the theory, laying the foundation for the Generalized Evolution Theory.
(*Note: The axiom numbering follows the original manuscript sequence. Axiom 3, focusing on the relativity of cognitive frameworks, is not elaborated here as it has low relevance to the core argument—see the complete manuscript of Noetic Ecology for details.*)

3.2 Defining Core Concepts: Ontological Distinctions Between Symbols, Information, and Noesis

To preempt terminological ambiguity, we emphasize that 'noetic' in this context specifically denotes systems capable of logical self-reference and self-organization—distinguishing them from passive information storage. This aligns with contemporary extended mind theory while extending it to civilizational scales.
To eliminate academic jargon and ensure clarity, we redefine symbols, information, and noesis here—emphasizing their dynamic differences:

- First, **Symbols**: As noetic genes (Axiom 4), symbols are the basic units of the noetic ecology and carriers of concepts, narratives, or meanings. Examples include the number "1," the mathematical operator "+", or the term "fairness." Symbols are static but can form complex structures through combination. Their core function is as hereditary units, replicated through education, imitation, and communication.

- Second, **Information**: Information refers to topological structures formed by the self-organization of symbols—arrangements, combinations, or relational networks of symbols following specific rules (e.g., logic, grammar, or cultural norms). Examples include the equation "1+1=2," a philosophical treatise, or a database. Information carries transmissible patterns but is relatively static, lacking active driving force; it requires external processing to be activated.
- Third, **Noesis**: Noesis denotes a complex information system endowed with logical self-reference (Axiom 0) and self-organization tendency (Axiom 2). Unlike static information, noesis is dynamic and self-driven: activated through information-processing media (e.g., brains or AI), it actively absorbs information, deconstructs (critical analysis) and reconstructs (innovative combination) it within the logical sub-universe, thereby driving the heredity, mutation, and selection of noetic genes. Examples include human consciousness, an active academic community, or a learning AI model. Noesis' defining trait lies in its intrinsic momentum—the continuous optimization drive sustained by the self-organization tendency, and its ability to simulate evolutionary processes free from spatiotemporal constraints, enabled by logical self-reference.

A critical distinction: Information serves as the "fuel" for noesis, but noesis endows information with meaning and direction through self-organization tendency and logical self-reference. This resolves the long-standing academic confusion between information and cognition.

3.3 A Systems Definition of Civilization: From Existential Foundations to Meaning Boundaries

Building on Axiom 1 (Material/Noetic Duality), we adopt an integrated systems definition of "civilization": Civilization is a macro-evolutionary system organized through complex noetic systems (information systems) on a material foundation. Typically unified by shared meta-narratives and collaborative protocols, it exhibits tendencies toward self-sustainment and expansion.

Two complementary dimensions shape this definition:

- Functionally (ontologically): Civilization is first and foremost a "material-noetic coupling system." Material entities (human bodies, cities, infrastructure) serve as its carrier and energy base, while noesis (technology, institutions, knowledge) acts as the key to its organizational patterns and functional realization. This dimension determines civilization's capacity boundaries and evolutionary potential.
- Culturally (topologically): Civilization is simultaneously a "narrative-protocol community." Shared core meta-narratives (e.g., cultural values, identity) and their derived collaborative protocols (e.g., laws, norms) define its internal sense of identity and external boundaries. This dimension determines civilization's cohesion and stability.

The relationship between the two dimensions:

- Meta-narratives themselves are the highest-level noetic genes, dependent on the underlying material-noetic architecture for survival and transmission.
- Conversely, the evolution of the material-noetic architecture (e.g., technological revolutions) can shake or reshape meta-narratives; powerful meta-narratives, in turn, can drive the directional development of the material-noetic architecture (e.g., the space race).

3.4 Formal Construction of the Generalized Evolution Model

Guided by the four core axioms above, we derive the Generalized Evolution Theorem and construct a formal model:

Here we arrive at Theorem 37, the Generalized Evolution Theorem:

Dependencies: Axiom 1 (Material/Noetic Duality), Axiom 2 (Driven by Will to Survive)

Statement: Evolution is a universal process through which complex systems achieve hierarchical leaps via mutation, selection, and inheritance. The evolution of the noetic ecology represents the natural continuation of biological evolution at the cognitive level; technology, institutions, and culture all qualify as evolvable "noetic phenotypes."

Significance: Unifies the evolutionary landscape from material to noetic realms, integrating technological change into the natural evolutionary lineage.

Corollary 37.1: The Dominance Law of Noetic Evolution

Statement: Once an intelligent system achieves internal simulation capacity through "logical self-reference" (Axiom 0), the primary driver and rate of its adaptive evolution shift from slow, random biological genetic mutation-selection to fast, directional noetic gene (symbol) mutation-selection. The observed phenomenon of "human evolution stagnation" is essentially a hierarchical leap in the main battlefield of evolution, not its termination. The yardstick for measuring civilizational evolution must therefore expand to include the complexity and synergy of the noetic ecology—not just the genetic diversity of populations in the material world.

Value:

- Direct engagement: Explicitly responds to observational facts in biology, offering a higher-dimensional explanation that withstands scrutiny.
- Transcendence: Establishes our theory as the natural continuation and inevitable transcendence of biological evolution at the civilizational scale.
- Legitimacy: Demonstrates that our theory is not a castle in the air but rooted in solid scientific facts, pointing the way forward.

(*Note: Theorem 37 is numbered based on its status as the 37th abstract insight of the author—not derived from 36 preceding theorems. We clarify this to avoid misperceptions about its logical dependencies.*)

Formalization of the model:

- **Noetic Gene Transmission Equation**: Describes changes in the prevalence of noetic genes in populations, highlighting their high-speed nature:

$$\frac{dP}{dt} = \beta \cdot I \cdot C \cdot P \cdot (1 - P) \cdot N - \delta P - \gamma P \cdot Q$$

Where: P = prevalence; β = infection intensity (degree to which the noetic gene satisfies the will to survive); I = influence distribution; C = framework compatibility; N = network effect coefficient; δ = forgetting rate; γ = competitive inhibition coefficient; Q = prevalence of competing genes. This equation quantifies the diffusion rate of noetic genes, which is far higher than that of biological genes.

We clarify the influence distribution (I): It refers to the relative weight of noetic gene transmitters in social networks. For example, academic authorities, key opinion leaders (KOLs), or top journals transmit genes with significantly higher I values than ordinary individuals. In empirical research, this parameter can be approximated using network centrality metrics (e.g., PageRank) or academic influence indices (e.g., H-index).

- **Cognitive Escape Model**: Describes the probability of individuals breaking through dominant paradigms, linked to civilizational phase transitions:

$$P(\text{escape}) = k \cdot \frac{M \cdot C_d}{A \cdot N_p}$$

Where: M = metacognitive ability; C_d = intensity of cognitive dissonance; A = paradigm authority; N_p = number of individuals internalizing the paradigm; k = cognitive mobility constant (influenced by education, technology, etc.). When P("escape") exceeds a critical threshold, a civilizational-level paradigm shift occurs.

Qualitative elaboration of the selection mechanism:

The selection mechanism in our model is not a vague analogy but maps to observable, concrete screening forces through equation parameters—resolving memetics' long-standing ambiguity:

- Social consensus selection: Manifested in parameters C (framework compatibility) and β (infection intensity). A noetic gene achieves high C only if compatible with the group's dominant cognitive frameworks and values ("social consensus"); it attains high β only if addressing the group's current survival anxieties or aspirations ("collective will"), thereby gaining widespread acceptance.
- Technological standard selection: Reflected in parameters δ (forgetting rate) and γ (competitive inhibition). In technology, a noetic gene (e.g., an algorithm or protocol) faces increased δ (elimination rate) or suppression by more efficient competing genes ($\gamma P \cdot Q$ term) if incompatible with dominant technical standards or unable to prove superior efficacy.
- Institutional selection: Embodied in N (network effect coefficient) and A (paradigm authority) in the Cognitive Escape Model. Formal and informal institutions—such as education systems, academic norms, and patent regimes—shape information flow networks and paradigm authority, thereby systematically screening and promoting the dissemination of certain noetic genes.

Thus, our model decomposes memetics' vague "selection" into operational, measurable dynamic parameters, transforming it into a testable scientific concept.

3.5 Positioning Relative to Existing Theories

- Transcending Memetics: Resolves memetics' ambiguous selection problem through clear noetic gene definitions and selection mechanisms (e.g., social consensus or technical standards).
- Integrating Technological Evolution Theory: Treats technology as a noetic phenotype, its evolution governed by noetic gene dynamics rather than an independent process.
- Materializing Systems Science: Converts abstract self-organization concepts into testable mechanisms through operational models (e.g., transmission equations and escape models).

The following table captures how we integrate and transcend existing approaches:

Theoretical Dimension	Memetics	Cultural Evolution Theory	Technological Evolution Theory	Complex Adaptive Systems Theory	This Study: Generalized Evolution Theory
Core Unit	Meme (vague)	Cultural Trait	Technological Component	Rule/Agent	Noetic Gene (symbolic structure)
Selection Mechanism	Vague, no clear analogy	Social Learning Biases (e.g., frequency dependence)	Market Selection, Efficacy Competition	Fitness Function (often exogenous)	Operationalized multiple selections: Social Consensus (C), Technical Standards (δ, γ), Institutions (N, A)
Mutation Mechanism	Random Misinterpreta tion	Innovation, Random Mutation	Combinatorial Evolution	Rule Crossing, Mutation	Open-Source Collaboration, Conceptual Recombination, Algorithmic Search
Heredity Mechanism	Imitation	Social Learning	Knowledge Accumulation, Blueprint Transmission	Rule Transmission	Education, Digital Replication, Protocol Inheritance
Relationship with Material Foundations	Disconnecte d	Disconnected	Implied, not unified	Emphasized, no clear duality	Integrated (Axiom 1: Material/Noetic Duality)
Intrinsic Driving Force	None	None	None	Adaptation	Integrated (Axiom 2: Self-Organization Tendency of Systems)
Applicable Scale	Cultural Domain	Cultural Domain	Technological Systems	Universal, macro-phenomenon-focused	Full-Scale Unification (Biological-Cultural-Technological)

Theoretical Dimension	Memetics	Cultural Evolution Theory	Technological Evolution Theory	Complex Adaptive Systems Theory	This Study: Generalized Evolution Theory
Quantifiability	No	Partial (models)	Partial (economic indicators)	Yes, often abstract simulation	Yes (provides concrete tools like transmission equations and escape models)
Core Limitation	Vague selection, ambiguous units	Isolated from biological/technological evolution	Ignores cognitive frameworks	Insufficient operability, hard to connect to specific cases	As an emerging framework, its quantitative predictive power awaits further empirical validation

3.6 Summary of This Chapter

Based on the axiomatic system of Noetic Ecology, this chapter constructs the theoretical framework of the Generalized Evolution Theory:

- Anchored in Material/Noetic Duality and the Self-Organization Tendency of Systems, it expands the scope of evolution to the noetic realm.
- Develops quantitative tools such as the Noetic Gene Transmission Equation and the Cognitive Escape Model.
- Clarifies the theory’s positioning relative to existing frameworks.

This framework provides a testable theoretical model for explaining the "accelerated evolution of human civilization." In the next chapter, we will use this model to elaborate on the specific mechanisms of civilizational acceleration.

4. Mechanism Demonstration: The Dynamics of Civilization’s Accelerated Evolution

Building on the theoretical framework established in Chapter 3, we now delve into the intrinsic mechanisms driving civilization’s accelerated evolution. We analyze four interconnected dimensions, blending qualitative reasoning with quantitative support to unpack this phenomenon:

4.1Mechanism 1: Strengthened Self-Organization Tendency and High-Speed Iteration of Noetic Genes

At the noetic level, the Self-Organization Tendency of Systems (Axiom 2) manifests as a powerful drive to maintain and expand ordered noetic structures—what we term "will to survive." This tendency has been significantly strengthened in contemporary society through two key pathways:

4.1.1 The Paradigm Shift in Will to Survive

Traditional survival pressures have shifted from individual physiological needs to civilizational competition. Falling behind in a single technological paradigm can condemn an entire civilization to marginalization in global competition (e.g., the long-term predicaments of nations that missed the Industrial Revolution). This systemic survival pressure translates into an institutional hunger for innovation: global R&D spending has surged from approximately $20 billion in 1960 to $2.5 trillion in 2023, with an annual growth rate far outpacing GDP.

4.1.2 Exponential Growth in Noetic Gene Mutation Rates

Supported by the External Brain Domain (digital networks), the mutation-selection cycle of noetic genes has been drastically compressed:

- Explosive growth in mutation sources: 8 billion globally connected individuals now serve as potential innovation nodes, generating billions of new information pieces daily.
- Intensified selection mechanisms: Market feedback, academic citations, and social media dissemination operate in real time, accelerating the weeding-out of inferior noetic genes.
- Enhanced genetic fidelity: Digital storage ensures nearly lossless transmission of noetic genes, preserving their core structure during replication.

We quantify this process with a model:

Effective Mutation Rate of Noetic Genes = $\mu\_base \cdot N \cdot k\_c$

Where N (number of connected individuals) has grown from millions in the 1970s to billions in the 2020s; k_c (connection efficiency coefficient) has improved exponentially with advances from dial-up internet to 5G/fiber optics. Their product—$N \cdot k\_c$—has increased by a factor of 10^4, a reasonable estimate based on communication technology history that fully explains the explosive growth in mutation potential.

4.2Mechanism 2: Catalytic Effects of the Civilizational Phase Transition Model

According to our civilizational dynamics framework, when the proportion of "Second-Order Noetic Humans" (individuals with metacognitive abilities) exceeds a critical threshold, the system undergoes a phase transition—shifting from a "Collective Mind" to "Metacognitive Collective Intelligence." This transition drastically boosts evolutionary rates.

4.2.1 Popularization of Education and Expansion of the Metacognitive Base

Global gross enrollment rate in higher education has risen from less than 1% in 1900 to over 40% in 2020, with emerging nations like China surpassing 50%. This expansion brings three key changes:

- Increased M (metacognitive ability): Systematic thinking training enables individuals to break free from traditional frameworks more easily.
- Sensitized C_d (cognitive dissonance): Education enhances individuals' ability to perceive contradictions within the system.
- Approaching critical mass: Complex systems theory suggests that new paradigms break through transmission thresholds when innovators account for 10-20% of the population.

4.2.2 Maturation of Collaborative Protocols

Modern academic norms, open-source collaboration, and patent systems form a set of efficient selection protocols for noetic genes:

- Truth-discovery protocols: Peer review and experimental replication screen for reliable knowledge.
- Attention-allocation protocols: Citation indices and impact factors optimize the distribution of cognitive resources.
- Value-alignment protocols: Research ethics and technical standards ensure innovation aligns with broader human interests.

These protocols significantly reduce the difficulty of cognitive escape ($D_c = (A^2 \cdot N_i) / M_a$) by boosting the denominator term, thereby increasing P(escape) (the probability of breaking through paradigms).

4.3Mechanism 3: Amplification Effects of Logical Self-Reference and the External Brain Domain

The combination of Logical Self-Reference (Axiom 0) and the External Brain Domain has created an unprecedented cognitive accelerator—revolutionizing how noetic genes mutate and iterate.

4.3.1 The Computational Revolution of the Logical Sub-Universe

Human individuals gain powerful cognitive extensions through the External Brain Domain:

- Simulation capabilities: Industry experience confirms that shifting from physical testing (e.g., wind tunnels for aerodynamics) to digital twin simulations (e.g., computational fluid dynamics) reduces costs (time, materials, labor) by a conservative factor of 1,000. This transformation has been validated in aerospace, automotive manufacturing, and other R&D-intensive fields.
- Expanded search spaces: AI algorithms can explore millions of design solutions in hours—far beyond human capacity.
- Accelerated iteration rates: We define iteration speed as $R_{iteration} = 1 / (T_{thinking} + T_{testing})$. Digital simulation drastically shortens $T_{testing}$ (validation time): for example, chip design via EDA software reduces each iteration from months of "manufacture-test" to hours, boosting $R_{iteration}$ by hundreds of times.

4.3.2 AI as a Cognitive Lever

Large language models and other AI technologies enable the directed evolution of noetic genes:

- Guided mutation: Prompt engineering steers innovation toward specific directions, avoiding random search.

- Preemptive selection: Multiple rounds of selection occur in virtual environments before physical implementation.
- Cross-domain recombination: AI breaks down traditional disciplinary boundaries, enabling conceptual "hybridization" across knowledge areas.

This "cognitive lever" effect can be quantified as L_AI = Productivity_AI / Productivity_human. In some fields, this coefficient exceeds 100: for instance, AlphaFold2 completes protein structure prediction tasks in days that might take human expert teams years—making L_AI far greater than 100. This metric aims to quantify AI's amplification of specific cognitive tasks.

4.4Mechanism 4: Reinforced Feedback Loop Between Cultural Evolution and Technological Innovation

Cultural evolution and technological innovation share a tight dynamic link, forming a closed reinforced feedback loop that serves as a key driver of civilizational acceleration.

4.4.1 Supply of Cognitive Frameworks: The Foundation of Cultural Evolution

Through education, cross-group exchange, and knowledge accumulation, cultural evolution builds a shared "logical sub-universe" and unified worldview. This cognitive framework provides individuals with thinking tools to transcend experiential boundaries, converging scattered individual cognition into systemic innovative potential—aligning with the self-organization tendency (Axiom 2) by expanding ordered information structures through both noetic gene iteration and cultural cognitive consensus.

4.4.2 Driving Blueprint Design: Translating the Logical Sub-Universe into Practice

Shared cognitive frameworks foster large-scale thought experiments and virtual simulations, deconstructing old problems and constructing new solutions within the cognitive space to form actionable "technical blueprints." This process synergizes with the External Brain Domain's computational support: digital tools further reduce the cost of blueprint design and accelerate the transformation from cognition to practical solutions.

4.4.3 Realizing Technological Materialization: From Blueprint to Innovation

Through engineering practice and industrial collaboration, technical blueprints are transformed into concrete tools, products, and socio-technical systems. This materialization process is both an explicit expression of noetic genes and a means for civilizational systems to develop new ways of interacting with matter and energy—strengthening the order and expansion capacity of self-organizing systems.

4.4.4 Closing the Loop: Cultural Feedback from Technological Innovation

New technologies (e.g., telescopes, the Internet, AI) act as "extended organs" for human perception and collaboration, significantly expanding observational boundaries and collaboration efficiency. They not only drive scientific discoveries (e.g., cosmology, micro-particle research) but also shape new cultural forms (e.g., digital culture, global collaborative culture)—injecting fresh knowledge into cultural evolution and enabling continuous iteration of cognitive frameworks.

4.5Synergistic Effects of Accelerating Mechanisms

These four mechanisms do not act in isolation but amplify each other through positive feedback loops:

4.5.1 Analysis of Reinforcement Loops

These mechanisms don't operate in isolation; they lock into a powerful, self-reinforcing cycle:

Strengthened self-organization tendency → Resource allocation toward innovation → Popularization of education and AI development
→ Enhanced metacognitive abilities and improved External Brain Domain
→ Increased cognitive escape probability and accelerated iteration rates
→ Faster mutation and selection of noetic genes
→ Growth in civilizational complexity
→ Further strengthened self-organization tendency (closed loop)

This virtuous cycle creates the engine of sustained acceleration we observe.

4.5.2 Quantitative Characterization: The Civilizational Potential Function

Based on the state function of civilizational dynamics, we observe:

Civilizational Potential = Σ(Individual Niche Realization) × System Resource Mobility × (1 - System Parasite Burden)

During the phase of accelerated evolution, all three factors rise simultaneously:

- Individual niche realization: Improves due to educational popularization and occupational differentiation.
- System resource mobility: Optimized by digitalization and globalization, facilitating resource allocation.
- System parasite burden: Relatively reduced through institutional innovation and technological progress.

4.6Evidence Support and Historical Comparison

4.6.1 Diachronic Comparison of Evolutionary Rates

- Biological evolution: The 1.2% genomic difference between humans and chimpanzees accumulated over 6-7 million years.
- Technological evolution: The transition from steam engines to internal combustion engines took ~150 years; from personal computers to smartphones, ~30 years.
- Algorithmic evolution: ImageNet image recognition accuracy rose from 72% in 2010 to over 99% in 2020.

4.6.2 Spatiotemporal Distribution of Innovation Density

Global patent grants have grown from ~500,000 per year in 1980 to ~3.5 million per year in 2020. Their distribution has expanded from traditional hubs (Europe, North America, Japan) to emerging innovation regions (China, India)—validating the global trend of cognitive escape.

4.7Conclusion of This Chapter

Through analysis of these four interconnected mechanisms, we confirm the core proposition of the Generalized Evolution Theory: human civilization is in an era of noetic-dominated accelerated evolution. This process is no random phenomenon but an inevitable outcome of the system's self-organization tendency under specific conditions.

In the next chapter, we will further validate the explanatory power of this theoretical framework through concrete case studies.

5. Case Studies: Empirical Evidence of Noetic Genes' Accelerated Evolution

A theory's value lies in its explanatory power. Here, we turn to two in-depth case studies to empirically examine the mutation-selection process of noetic genes in contemporary environments. The first focuses on the technological dimension, tracing the evolution of the artificial intelligence (AI) tech stack as a paradigmatic noetic gene. The second centers on the cultural sphere, analyzing the global dissemination of the concept "involution" as a cultural gene. Together, they form the empirical cornerstone of the Generalized Evolution Theory.

5.1 Case Study 1: The Evolution of the AI Tech Stack

The development of AI offers an almost perfect observational window into the high-speed mutation and selection of noetic genes.

5.1.1 Identifying Noetic Genes: Deep Learning Architectures as Core Carriers

We identify deep learning neural network architectures as the core noetic genes in the AI domain. Their basic structure (hierarchical transformation, backpropagation, weight adjustment) forms the genetic core, while specific implementations (CNN, RNN, Transformer, etc.) represent mutant phenotypes.

- Genetic stability: Backpropagation, explicitly formulated by Rumelhart et al. (1986), has persisted as the core algorithmic paradigm.
- Mutation activity: Architectural details evolve rapidly—from AlexNet's 8 layers in 2012 to GPT-4's approximately one trillion parameters in 2023, complexity has grown exponentially.

5.1.2 Mutation Mechanisms: Open-Source Ecosystems and Algorithmic Recombination

Noetic gene mutation in AI unfolds through three key pathways:

- Parallel exploration in open-source communities
  Platforms like GitHub serve as global "mutation laboratories." Take the Transformer architecture: since its paper publication in 2017, it has spawned dozens of major variants (BERT, GPT, T5, etc.). Tracing relevant open-source projects on GitHub reveals that key variants emerge at an average interval of just 3-4 months (Source: GitHub Trending Repositories, 2018-2023)—a testament to the extremely high mutation rate of noetic genes.
- Cross-domain recombination of algorithmic components
  The fusion of CNN (from computer vision) and Attention mechanisms (from natural language processing) in Transformers yielded breakthrough results. This "conceptual hybridization," analogous to horizontal gene transfer in biology, drastically accelerates functional innovation.
- Directed evolution in hyperparameter space
  Automated machine learning (AutoML) explores architectural spaces beyond human intuition through neural architecture search, powered by thousands of GPU-days of computing power—uncovering highly efficient network structures.

(A comprehensive analysis of the top 100 AI-related repositories on GitHub from 2018-2023 confirms this 3-4 month variant emergence cycle.)

5.1.3 Selection Environment: Dual Selection Pressures

AI noetic genes face two layers of selection criteria:

- Technical efficacy selection: Objective metrics like accuracy and speed on standard datasets form the first filter.
- Application ecosystem selection: Industry adoption, developer willingness, and computing costs constitute the second filter.

As a macro manifestation of this high-speed mutation-selection cycle, AI model complexity has grown exponentially. According to OpenAI's research, computing power required for large-scale AI training has increased 10-fold every ~18 months (Source: OpenAI, "AI and Compute," 2018)—a trend consistently validated in subsequent *AI Index Reports*. Model parameters have surged from 60 million in AlexNet (2012) to approximately one trillion in GPT-4 (2023)—a more than 5-order-of-magnitude increase, far outpacing biological evolution.

5.1.4 Quantifying Evolutionary Rates

The update rate of noetic genes in AI is striking:

- Intervals between major architectural innovations have shrunk from 10-15 years (perceptrons to backpropagation) to 1-2 years in recent years.
- Top-5 error rates on ImageNet plummeted from 28.2% in 2010 to 1.5% in 2020—more than an 18-fold improvement in a decade.
- Model parameters grow exponentially, doubling every ~18 months (10-fold increase per cycle).

This evolutionary rate is 5-6 orders of magnitude faster than traditional evolution based on biological genes.

5.2 Case Study 2: The Global Dissemination of "Involution"

As a cultural noetic gene, "involution" (Chinese: 内卷) offers a clear trajectory of how conceptual mutation interacts with social selection.

5.2.1 Tracing the Noetic Gene: Origin and Mutation

Let's trace its evolutionary path:

- Academic origin (1963): American anthropologist Clifford Geertz described the phenomenon of intensive rice farming with diminishing marginal returns in Java in *Agricultural Involution*.
- Disciplinary migration (1980s-2000s): Scholars like Huang Zongzhi introduced the concept to Chinese social and economic history, describing "growth without development."
- Popular mutation (2018-2020): Chinese social media users redefined it to mean "irrational internal competition."
- Global diffusion (2020-2022): Through media coverage and academic exchanges, the concept entered the English-speaking world as "involution."

During dissemination, its core meaning underwent significant mutation:

- From systemic description (agricultural/economic system traits) to individual experience (personal competitive stress).
- From objective analytical tool to subjective emotional expression.

5.2.2 Analyzing the Selection Environment

The concept's successful dissemination stems from its deep resonance with specific social contexts:

- Structural social pressures: Popularization of higher education has devalued academic credentials; slowing economic growth has intensified job competition—providing a realistic foundation for the concept.
- Emotional resonance needs: It offers a concise, powerful label for young people's anxiety and exhaustion, meeting the demand for emotional expression in the social media era.
- Explanatory advantage: Compared to traditional terms like "fierce competition," "involution" more accurately captures the paradox of "increased input without increased gain."

5.2.3 Validating Dissemination Dynamics

We can semi-quantitatively verify this process using the Noetic Gene Transmission Equation:
During its peak dissemination in 2020:

- Infection intensity (β) was extremely high: The concept directly addressed the survival anxiety of contemporary youth.
- Framework compatibility (C) was strong: It aligned closely with China's "competitive culture" and "education anxiety."
- Network effect (N) was powerful: Social media enabled exponential dissemination.

Baidu Index data shows that search volume for "involution" (内卷) surged more than 100-fold between September and December 2020—validating the equation's prediction of explosive growth.

5.2.4 Selection Pressures in Globalization

When the concept spread to the English-speaking world (as "involution"), it faced different selection pressures:

- Competition with existing terms like "burnout" and "overwork."
- Need to adapt to Western individualistic cultural frameworks.

Selection outcome: It gained traction in academic and media circles but failed to replicate its mainstream popularity in the Chinese-speaking world—reflecting how different cultural environments select for the same noetic gene.

5.3 Comparing Cases and Theoretical Insights

Though distinct in domain, the two cases share key traits of noetic genes:

- The three-stage process: Both follow the basic evolutionary logic of mutation-selection-inheritance.
- Environment-specific selection: Technical efficacy vs. social resonance serve as domain-specific selection criteria.
- Dependence on communication networks: Open-source communities and social media act as critical dissemination channels.
- Accelerated evolution: Both exhibit evolutionary rates far exceeding biological evolution benchmarks.

Their differences are equally revealing:

- Divergent selection criteria: AI technology is dominated by objective technical metrics; cultural concepts are driven by subjective experience and social resonance—indicating domain-specific selection for noetic genes.
- Varied mutation constraints: AI technology is strictly bounded by mathematical and physical laws; cultural concepts are constrained by logical consistency and psychological acceptability—

constraints that define the potential space for mutation.

- Dissemination fidelity: AI technology (code, model parameters) can be replicated precisely; cultural concepts inevitably undergo semantic drift during transmission—affecting the long-term stability of noetic genes.

5.4 Theoretical Value of the Case Studies

These two cases provide solid support for the Generalized Evolution Theory:

- Confirm core hypotheses: Noetic genes indeed evolve at rates far exceeding biological genes.
- Validate the theoretical framework: The Noetic Gene Transmission Equation effectively describes real-world dissemination processes.
- Reveal mechanism details: Clarify the specific mutation and selection mechanisms of noetic genes across domains.
- Offer research methods: Demonstrate how tracking the dissemination of specific noetic genes can illuminate civilizational evolution.

The cases show that framing AI technological development and cultural concept dissemination as evolutionary processes is not only theoretically consistent but also possesses strong explanatory and predictive power.

6. Discussion and Conclusion: Theoretical Boundaries and Civilizational Insights of the Generalized Evolution Theory

After five chapters of theoretical construction and empirical validation, this paper now stands on solid ground to examine the full scope of the Generalized Evolution Theory. In this final chapter, this chapter addresses potential criticisms, clarifies theoretical boundaries, summarizes core contributions, and reflects on the profound insights this theory offers for understanding humanity's civilizational future.

6.1 Re-examining Core Concepts: Why Noetic Genes Are Not Memes

This section opens with a critical distinction that anchors the theory's uniqueness:
Core Argument:

- From vague to clear selection mechanisms: Memetics treats "selection" as a black box, while this study operationalizes it as measurable dynamic parameters—social consensus (C), technical standards (δ, γ), and institutions (N, A).
- Integration with material foundations: Memetics floats disconnected from physical reality; noetic genes establish an ontological link with biological evolution and technological hardware through "Material/Noetic Duality."
- Presence of intrinsic driving force: Memes spread passively; noetic gene evolution is actively driven by the "Self-Organization Tendency of Systems" (Axiom 2), endowing it with directionality.
- Clear hierarchical structure: Memes are a flat concept; noetic genes form a layered architecture—symbols (atoms) → information (molecules) → noesis (active systems).

This distinction resolves the long-standing ambiguity of memetics, grounding the theory in concrete, testable mechanisms.

6.2 Theoretical Boundaries and Addressing Criticisms

No scientific theory is complete without clarifying its scope and confronting objections. **This section addresses four key criticisms of the Generalized Evolution Theory**:

6.2.1 Ontological Criticism: Does Noesis Possess Independent Reality?

Criticism Core: Treating "noesis" as a fundamental category alongside "matter" smacks of dualism. Is noesis not merely a complex manifestation of matter?

Response:

The Generalized Evolution Theory is built on Axiom 1 (Material/Noetic Duality)—where 'noetic' captures the emergent, self-organizing cognitive-informational dynamics in complex systems—extending Clark and Chalmers' (1998) 'extended mind' thesis to civilizational scales
, not traditional dualism but a systems science perspective.

It acknowledges:

- Ontological priority: Matter is ontologically primary; noesis emerges from specific material organizations.
- Causal efficacy: Once emergent, the noetic layer follows its own dynamic laws and exerts downward causal effects on the material layer.

- Theoretical necessity: Just as biology cannot be reduced to physics, noetic layer laws cannot be fully reduced to material layer processes.
- This conceptualization of noetic systems builds upon while extending Clark and Chalmers' (1998) 'extended mind' thesis to civilizational scales, providing a unified framework for understanding how cognitive processes operate across biological and technological substrates.

This stance aligns with mainstream views in contemporary complexity science and systems philosophy.

6.2.2 Methodological Criticism: Is Analogy Equivalent to Mechanism?

Criticism Core: Applying biological evolution concepts (mutation, selection, inheritance) to the noetic realm is merely a loose analogy, not a rigorous scientific mechanism.

Response:

The Generalized Evolution Theory transcends metaphor to establish operational mechanisms:

- Mutation mechanisms: Realized through open-source collaboration, conceptual recombination, and algorithmic search.
- Selection mechanisms: Operate through clear criteria like technical standards, market adoption, and academic evaluation.
- Inheritance mechanisms: Occur via concrete channels such as education, code replication, and citation of literature.

These mechanisms are detailed in the case studies and semi-quantitatively described through tools like the Noetic Gene Transmission Equation.

6.2.3 Empirical Criticism: Is Accelerated Evolution Falsifiable?

Criticism Core: Can the proposition of accelerated civilizational evolution be falsified? Are there clear counterexamples?

Response:

The Generalized Evolution Theory makes several testable predictions (assuming basic civilizational stability):

- Prediction 1: Innovation rates in key fields (e.g., AI, biotechnology) will maintain or exceed current levels over the next decade.
- Prediction 2: Innovation metrics like global patents and paper outputs will continue superlinear growth.
- Prediction 3: A significant positive correlation will persist between educational attainment and regional innovation efficacy.

Sustained technological stagnation, long-term declines in innovation metrics, or decoupling of education and innovation would constitute valid counterevidence.

6.2.4 Axiological Criticism: Does Accelerated Evolution Imply Progress?

Criticism Core: Equating evolutionary acceleration with positive development commits a "progress fallacy." Could acceleration bring systemic risks?

Response:

The theory adopts a strictly value-neutral stance on "progress." "Acceleration" here is a dynamic descriptor—like stating an object is speeding up—carrying no moral judgment of "good" or "bad." Accelerated evolution is a natural outcome of the noetic ecology’s internal logic: it amplifies both civilizational creativity and destructiveness. **The theory’s purpose is to reveal this objective law, providing a cognitive foundation**

for guiding and harnessing this accelerating force.

6.3 Summarizing Core Contributions

Against the backdrop of addressing criticisms, **the core contributions of the Generalized Evolution Theory are fourfold**:

6.3.1 Paradigmatic Shift: From Narrow to General Evolution

The theory pioneers three leaps in evolutionary thinking:

- From biology to culture: Extending evolutionary logic to the cultural realm.
- From culture to technology: Integrating technological systems into evolutionary analysis.
- From separation to unification: Establishing a unified framework for biological, cultural, and technological evolution.

This paradigmatic shift allows explaining phenomena from genes to memes to technology within a coherent theoretical framework.

6.3.2 Conceptual Innovation: Operationalizing Noetic Genes

Through the core concept of "noetic genes," **the study achieves three key outcomes**:

- Resolves memetics' ambiguous unit definition problem.
- Establishes a cross-domain analytical unit for comparative research.
- Lays a conceptual foundation for quantifying civilizational evolution.

This innovation fills an analytical gap in evolutionary theory for cultural and technological domains.

6.3.3 Clarifying Mechanisms: Materializing Mutation-Selection Processes

Through case studies and equation descriptions, **the theory specifies three core mechanisms of noetic gene evolution**:

- Mutation: Open-source innovation, conceptual recombination, algorithmic search.
- Selection: Efficacy standards, social resonance, institutional screening.
- Inheritance: Educational transmission, technological diffusion, literary citation.

These mechanisms elevate the theory from metaphor to testable scientific hypotheses.

6.3.4 Enabling Prediction: Scientific Tools for Civilizational Futures

Based on the civilizational dynamics model, **the theory provides new tools for foresight**:

- Predicting paradigm shifts via cognitive escape probability.
- Modeling innovation diffusion through the Noetic Gene Transmission Equation.
- Diagnosing system health using the civilizational state function.

These tools offer a scientific basis for consciously guiding civilizational development.

6.4 Insights for Civilizational Futures

The Generalized Evolution Theory not only helps understand the past but also illuminates humanity's future path:

6.4.1 Redefining Humanity's Evolutionary Mission

If noetic evolution has become the dominant mode, humanity's evolutionary mission has fundamentally shifted:

- From adapting to the environment to shaping it: Humans have become the primary shapers of Earth's ecosystem.
- From passive evolution to active guidance: Directing evolutionary trajectories through technological choices and institutional design.
- From individual optimization to system optimization: Shifting focus from individual fitness to systemic sustainability.

This redefinition carries profound implications for education, ethics, and policy-making.

6.4.2 Warning of Systemic Risks in Accelerated Evolution

The theory helps identify emerging risks from acceleration:

- Cognitive divides: Evolutionary rate disparities leading to understanding gaps between groups.
- Control dilemmas: Technological complexity outpacing direct human oversight.
- Meaning crises: Excessive instrumental rationality eroding existential purpose.
- Ecological overshoot: Technological systems imposing unsustainable pressure on natural systems.

These risks demand corresponding early warning and governance mechanisms.

6.4.3 Guiding Civilizational Health Diagnostics

Building on Axiom 5 (System Value Criterion), **the study proposes three core dimensions for diagnosing civilizational health**:

- Individual realization: Does the system promote the development of each member's potential?
- Innovation mobility: Does the system maintain sufficient mutation capacity and selection efficacy?
- Risk resilience: Can the system withstand risks from accelerated evolution?

This provides a value compass for evaluating and optimizing civilizational development paths. Its ultimate criterion stems from the theory's cornerstone—Axiom 5's empirical law: The survival of any noetic system depends on its ability to continuously meet the needs of its internal components (individuals). A system that betrays this principle will inevitably suffer declining health and stability.

6.5 Research Prospects and Future Directions

The Generalized Evolution Theory opens new avenues for inquiry:

6.5.1 Theoretical Deepening

- Mathematical development: Refining the Noetic Gene Transmission Equation into a more precise predictive model.
- Experimental validation: Designing controlled experiments to test noetic gene mutation-selection mechanisms.
- Cross-cultural comparison: Investigating noetic evolution pattern differences across cultural contexts.

6.5.2 Application Expansion

- Innovation prediction: Developing technological trend forecasting tools based on the theory.
- Educational design: Optimizing education systems according to noetic evolution laws.
- Policy evaluation: Establishing an evolution-inspired policy impact assessment framework.

6.5.3 Philosophical Reflection

- Nature of consciousness: Exploring the link between logical self-reference and conscious emergence.

- Value foundations: Reconstructing ethical systems in the context of accelerated evolution.
- Existential meaning: Reconsidering human uniqueness and value amid noetic evolution.

6.6 Final Conclusion

Returning to the core question that opened this paper: Has human evolution stagnated? Based on the comprehensive analysis of the Generalized Evolution Theory, **a clear negative answer emerges**.

Human evolution has not stagnated—it has undergone a battlefield shift and a rate leap. As noetic evolution became the dominant mode, human civilization entered an era of unprecedentedly accelerated evolutionary change. This transformation brings unparalleled opportunities and profound systemic challenges.

The value of the Generalized Evolution Theory lies not only in explaining the past but also in providing a new theoretical compass for navigating the future. It tells us that civilization's fate is no longer determined by blind natural selection but increasingly by humanity's ability to understand and guide its own noetic evolution.

In this sense, the Generalized Evolution Theory is more than a scientific theory—it is a call for civilizational awareness. It invites humanity to shape the future of itself and the planet with greater consciousness and responsibility—a gift, and perhaps the greatest test, bestowed upon our species by noetic evolution.

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